1 Notation

We'll investigate the generating functions for quarter-plane walks where we track how many times these walks touch the walls.

- Q is the generating function.
- Powers of variables x_i keep track of the i-th coordinate of the endpoint of a random walk. For example the monomial $x_1^3 x_2^1 x_4^7$ counts how many walks end at $(3, 1, 0, 7)$.
- Powers of variables a_i keep track of the number of times a random walk passes through a point with i-th coordinate equal to zero. For example the monomial $a_1^2 a_2$ counts how many times the random walks have passed through points with coordinates $(0, j)$ and $(i, 0)$ respectively.
- Powers of t count the length of random walks.
- Variables with overline are reciprocals like $\overline{x_1} = 1/x_1$.
- S is the Laurent polynomial generating function for the allowed step set. For example for the steps {N, E, SW} we have $S = x_2 + x_1 + \overline{x_1 x_2}$.
- We'll often use $b_i = a_1 - 1$ and $c_i = b_i/a_i = 1 - \overline{a_i}$.

Next we have some notation on extraction of coefficients

- $[d] = (1, \ldots, d)$ as a sequence.
- For a sequence $I = (i_1, \ldots, i_n)$ and a sequence $P = (p_1, \ldots, p_n)$, $x_I^P = x_{i_1}^{p_1} \ldots x_{i_n}^{p_n}$. For example $x_{[3]}^{(1,0,2)} = x_1^1 x_2^0 x_3^2 = x_1 x_3^2$. We'll usually use this notation for when we have general sequences, not in the case of the example which we would write out explicitly.
- $[u_I^P]$ stands for extracting the coefficient infront of u_I^P in the generating function expansion.
- $Q_I^J = [x_I^P]Q$ and similarly $S_I^J = [x_I^J]S$. For example if $I = (1, 3)$, $J = 0, 1$, (omiting parentheses) $Q_I^J = Q_{1,3}^{0,1}$. We will sometimes write $(Q_I^P)_J^R = Q_{I,J}^{P,R}$
- For a set I, $\mathbb{O}$ and $\mathbb{I}$ stand for sequences of $|I|$ zeros and ones respectively. For example for $I \subseteq [d]$ we might have $S_I^{-\mathbb{I}} Q_I^{\mathbb{O}}$. Analogously we will write $(Q_I^{\mathbb{I}})_J^{\mathbb{O}} = Q_{I,J}^{\mathbb{I},\mathbb{O}}$

So we have

$$Q(x_1, x_2, a_1, a_2; t) = \sum_{i,j,k} Q_{1,2}^{i,j} x_1^i x_2^j t^k$$

So far we won't extract a's because we have an unobvious choice between extracting a, b or c with various pros/cons. For example it may sometimes be easier to extract b_1^k than a_1^k but there's the added cost that the extracted term counts walks with varying powers of k. We usually won't extract with respect to t.

2 Introduction of Methods - 1D random walks

2.1 Derivation of the functional equation

In the interest of working out higher dimensional labelled walks' problems it is very useful to first visit the 1-dimensional case which will illustrate a lot of the methods that will later come into play. The 1-dimensional case will also give us a clearer picture as to how things work in higher dimensions (although the picture becomes more complicated). We'll treat the problem using generating functions. We'll use lower case letters and not quite the full notation because it is not necessary. In one dimension, on one axis, we'll use steps of length at most 1. If we have a random walk ending at coordinate i we can either step back adding -1 to the coordinate, step forward adding $+1$ to the coordinate or staying in one place, adding 0 to the coordinate. This can be somewhat formalized by using polynomials. Ending a walk at coordinate i will be encoded with a monomial x^i. Then stepping backwards, forwards or staying in one place will be denoted with x^{i-1}, x^i and x^{i+1} respectively. This is the same as multiplying by $1/x = \overline{x}, 1, x$ in each case. We can encode this information is a Laurent step polynomial s defined as

$$s = s_{-1}\overline{x} + s_0 + s_1 x$$

Here s_i will be either 0 or 1 for any specific i, tracking whether we have allowed those kinds of steps or not. Keeping this in mind, there are 8 possible step sets because we have two options for each s_i. We keep track of the length of a random walk with powers of t and we keep track of its endpoint with powers of x. We'll also label every interaction with the origin by multiplying by a every time a walk steps on the origin. We'll keep the walks on the non-negative axis.

Long story short we can find the generating function in a very simple, brute force manner, even for higher dimensions. We start with the empty walk denoted by 1 since it starts and ends on the origin without taking a step, so no t, x or label a. Then try out each step from s and eliminate those that take us past the origin in the negative direction. If we step on the origin we multiply by a. Those walks that remain are then multiplied by t since they are not of length 1. We can proceed iteratively k tiems to obtain random walks on the non-negative axis of length k. Adding up the results allows us to write the generating function up to $O(t^k)$. It will prove very useful to think about operators so we can formalize this notion in terms of operators. For example, here we are taking walks of length $k-1$, trying all options of s (this is like multiplying by s), removing all the walks that end up on the negative axis (this is like removing the $\overline{x}$ term) and labeling those that end on the origin (this is like multiplying the x^0 coefficient by a). Let D stand for the operator that cleans up the walks and labels the valid ones. Then if q_k stands for the labelled walks of length k, then the k-th term of q is $q_k t^k$ and we have

$$q_{k+1}t^{k+1} = tD(sq_k t^k)$$

Thus we can write the generating function q iteratively

$$q = \sum_{k=0}^{\infty} t^k (Ds)^k (1)$$

where we're iteratively applying Ds to 1 to obtain all labelled walks. We can treat this operator Ds as a formal variable and write

$$q = \sum_{k=0}^{\infty} (tDs)^k(1) = \frac{1}{1 - tDs}(1)$$

and then this gives us a functional equation for q

$$(1 - tDs)q = 1$$

There is another way of deriving this equation while not treating Ds as a formal variable. Instead of applying the operator iteratively we can apply it on the generating function as a whole. The product tsq is the generating for all random walks that end with x-coordinate greater or equal to -1 where some steps end up on x-coordinate 0 without being labelled. We can apply D on this product to remove the terms ending at x-coordinate -1 and label those that end up on x-coordinate 0. This leaves us with all the valid walks encoded in the generating function q without the empty walk 1. Therefore we can write a functional equation for q as

$$q = 1 + D(tsq) = 1 + tDs(q)$$

and we ca re-write this as $(1 - tDs)q = 1$ as before. Now, we have to say a few words about what the operator D actually is. First we are removing all the walks that end on x-coordinate equal to -1 which can be handled by the operator $E = 1 - \overline{x}[\overline{x}]$ which removes those walks. Then the labelling can be handled by the operator $L = 1 + (a - 1)[x^0] = 1 + b[x^0]$ where $b = a - 1$. Therefore we can write $D = LE$

$$\begin{aligned} D = LE &= (1 - \overline{x}[\overline{x}])(1 + b[x^0]) \\ &= 1 + b[x^0] - \overline{x}[\overline{x}] - \overline{x}[\overline{x}]b[x^0] \\ &= 1 + b[x^0] - \overline{x}[\overline{x}] \end{aligned}$$

since the last term vanishes on all generating functions. We could have also applied the operators in reverse to get the same result. Thiscommutativity makes sense because on the level of the function they are operator interacting with different monomials. We can now expand the functional equation to the following

$$\begin{aligned} 1 &= q - tLE(sq) = q - t(sq + b(sq)_0 - \overline{x}(sq)_{-1}) \\ &= q - t(sq + b(s_{-1}q_1 + s_0q_0 + s_1q_{-1}) - \overline{x}(s_{-1}q_0)) \\ &= q - t(sq + b(s_{-1}q_1 + s_0q_0) - \overline{x}(s_{-1}q_0)) \end{aligned}$$

since $q_{-1} = 0$. We can rewrite this equation in the following form

$$(1 - ts)q - t(bs_0 - \overline{x}s_{-1})q_0 - bts_{-1}q_1 = 1$$

Here $q_0 = q(0)$ and $q_1 = q'(0)$ where $q = q(x)$. In the interest of simplifying this equation we

can eliminate the q_1 term by extracting the x^0 coefficient from the whole equation

$$\begin{aligned}
1 &= q_0 - t(s_{-1}q_1 + s_0q_0 + s_1q_{-1}) - tbs_0q_0 - bts_{-1}q_1 \\
&= q_0 - t(s_{-1}q_1 + s_0q_0) - tbs_0q_0 - bts_{-1}q_1 \\
&= (1 - ts_0 - tbs_0)q_0 - (ts_{-1} + bts_{-1})q_1 \\
&= (1 - ats_0)q_0 - t(s_{-1} + bs_{-1})q_1 \\
&= (1 - ats_0)q_0 - ats_{-1}q_1
\end{aligned}$$

We can solve this equation for q_1

$$ts_{-1}q_1 = \overline{a}((1 - ats_0)q_0 - 1)$$

and substitute the result into the functional equation (for now we don't worry about the possibility of $s_{-1} = 0$ because it leads to the same thing anyway).

$$\begin{aligned}
1 &= (1 - ts)q - t(bs_0 - \overline{x}s_{-1})q_0 - c((1 - ats_0)q_0 - 1) \\
&= (1 - ts)q - (c(1 - ats_0) + bts_0 - t\overline{x}s_{-1})q_0 + c \\
&= (1 - ts)q - (c - bts_0 + bts_0 - t\overline{x}s_{-1})q_0 + c \\
&= (1 - ts)q - (c - t\overline{x}s_{-1})q_0 + c
\end{aligned}$$

and therefore we can rewrite the equation as

$$(1 - ts)q - (c - t\overline{x}s_{-1})q_0 = \overline{a}$$

since $1 - c = 1 - (1 - \overline{a}) = \overline{a}$. This process is rather complicated, even though it can be done for higher dimensional problems as well (as you can see in various previous approaches). Also there is an issue that we need to consider every time we are solving an equation in order to substitute quantities back into the equation. This is where the operator notation comes to the resque. It turns out the the L operator is invertible with inverse given by

$$L^{-1} = M = 1 - c[x^0]$$

and we can apply it on the functional equation. The RHS is simply $M(1) = 1 - c = \overline{a}$ and the LHS is

$$\begin{aligned}
M(1 - tLEs)q &= Mq - tEsq \\
&= (q - cq_0) - t(sq - \overline{x}s_{-1}q_0) \\
&= (1 - ts)q - (c - t\overline{x}s_{-1})q_0
\end{aligned}$$

and we've derived the same equation but in a much neater way. This will prove crucial later on when we derive a functional equation for higher dimensional problems. Although it can be done directly with a lot of effort it's much better to just use the operator notation. The derivation is brought down from about a dozen pages to one or two. The operator notation will also allow us to solve some multidimensional problems directly or with the help of lower dimensional problems.

We can extract the x^0 coefficient from the final functional equation again and although we can do this directly, we'll use the operator notation to tidy up the process. This will show up later when dealing with higher dimensional problems

$$[x^0]M = [x^0] - [x^0]c[x^0] = [x^0] - c[x^0] = \overline{a}[x^0]$$
$$[x^0]E = [x^0] - [x^0]\overline{x}[\overline{x}] = [x^0]$$

Then we extract the x^0 from the functional equation

$$[x^0]\overline{a} = \overline{a} = [x^0](M - tEs)q = ([x^0]M - t[x^0]Es)q = (\overline{a}[x^0] - t[x^0]s)q = [x^0](\overline{a} - ts)q$$

Therefore the extraction leads to

$$[x^0](1 - ast)q = 1$$

We could extract higher coefficients from the functional equation too. Let $k \geqslant 1$. Then

$$[x^k]M = [x^k] - [x^k]c[x^0] = [x^k]$$
$$[x^k]E = [x^k] - [x^k]\overline{x}[\overline{x}] = [x^k]$$

Therefore

$$[x^k]\overline{a} = 0 = [x^k](M - tEs)q = ([x^k]M - t[x^k]Es)q = ([x^k] - t[x^k]s)q = [x^k](1 - ts)q$$

That is

$$[x^k](1 - ts)q = 0$$

and we can readily extract whichever coefficients we want. The first equation leads to

$$(1 - ats_0)q_0 - ats_{-1}q_1 = 1$$

and the second leads to

$$q_k - t(s_{-1}q_{k+1} + s_0q_k + s_1q_{k-1}) = 0$$
$$(1 - s_0t)q_k - t(s_{-1}q_{k+1} + s_1q_{k-1} = 0$$

2.2 Solving the functional equation for various s

In this section we'll solve all 8 one-dimensional problems fully. Some cases are easier to deal with than others. As we'll see later in the higher dimensional cases, some problems are easierto solve than others. This difficulty tends to scale up with the inclusion of certain steps with negative direction.

2.2.1 The $s_{-1} = 0$ case.

This condition allows us to solve explicitly for q_0.

$$q_0 = \frac{1}{1 - as_0t}$$

We can substitute this expression in the functional equation and obtain

$$q = \frac{\overline{a}}{1 - st}\left(1 + \frac{b}{1 - as_0t}\right) = \frac{1 - s_0t}{(1 - st)(1 - as_0t)}$$

This takes care of half of the cases! What happens when we set $a = 1$, corresponding to setting $b = 0$ or extracting the b^0 coefficient from the equation? This corresponds to unlabelling the random walks. We obtain

$$[b^0]q = \widehat{q} = \frac{1 - s_0t}{(1 - st)(1 - s_0t)}$$

We'll note that

$$(1 - as_0t)q - (1 - s_0t)\widehat{q} = 0$$

This can also be written in operator form with $B = 1 - [b^0]$

$$B((1 - as_0t)q) = 0$$

This relates the labelled generating function to the unlabelled generating function! it says we can obtain the labelled generating function from the unlabelled one by the following relation

$$q = \frac{1 - as_0t}{1 - s_0t}\widehat{q}$$

We can also establish such as relationship for q_0 although the derivation here will be trivial. Extracting $[x^0]$ from the equation leads to

$$B((1 - as_0t)q_0) = 0$$

and

$$q_0 = \frac{1 - as_0t}{1 - s_0t}\widehat{q_0}$$

The relationship between q and $\widehat{q}$ is no accident but it tends to happen fairly often in the higher dimensional cases that we've considered. The functional equation in the $s_{-1} = 0$ case is easier to solve most likely because when $s_{-1} = 0$, walks don't go beyond the origin in the negative direction. Consequently the E operator acts as the identity on the generating function. and the functional equation reads as

$$(M - ts)q = q - cq_0 - tsq = (1 - ts)q - cq_0 = \overline{a}$$

2.2.2 The $s_{-1} \neq 0$ case.

The functional equation in the $s_{-1} \neq 0$ is not as easy to solve so it requires more treatment. We'll try to apply the kernel method whenever possible. It requires finding an x value that makes the kernel $1 - ts$ equal to zero in the functional equation

$$(1 - ts)q - (c - t\overline{x})q_0 = \overline{a}$$

Such a solution will exist since $s_{-1} = 1$ and therefore the equation $1 - ts = 0$ is non-trivial. Let one such solution be $x = \sigma$ and suppose that we can plug it into the functional equation. Then the kernel vanishes, taking q away with it and we obtain

$$-(c - t\overline{\sigma})q_0 = \overline{a}$$

We can solve for q_0 explicitly again

$$q_0 = \frac{\overline{a}}{\overline{\sigma}t - c}$$

We can now solve for q

$$q = \frac{\overline{a}}{1 - ts}\left(1 - \frac{t\overline{x} - c}{t\overline{\sigma} - c}\right) = \frac{t}{1 - ts}\left(\frac{\overline{x} - \overline{\sigma}}{b - a\overline{\sigma}t}\right)$$

We note that we can apply the B operator again

$$B((a\overline{\sigma}t - b)q) = 0$$

or written out explicitly

$$(a\overline{\sigma}t - b)q - (\overline{\sigma}t)\widehat{q} = 0$$

This also allows us to write the labelled generating function in terms of the unlabelled generating function

$$q = \frac{\overline{\sigma}t}{a\overline{\sigma}t - b}\widehat{q} = \frac{t}{at - b\sigma}\widehat{q}$$

We can also work with q_0 at this stage. We find

$$[x_0]B((a\overline{\sigma}t - b)q) = B((a\overline{\sigma}t - b)q_0) = 0$$

too. This ends up being the same as

$$(a\overline{\sigma}t - b)q_0 - (\overline{\sigma}t)\widehat{q_0} = 0$$

with solution

$$q_0 = \frac{t}{at - b\sigma}\widehat{q_0}$$

The final task for this section will be to find what σ will be for various cases. We already know $s_{-1} = 0$ so the result depend on s_0 and s_1. In general the kernel is zero when

$$\begin{aligned} ts &= 1 \\ t(s_{-1} + s_0x + s_1x^2) &= x \\ ts_1x^2 - (1 - s_0t)x + ts_{-1} &= 0 \end{aligned}$$

When $s_1 \neq 0$ this is quadratic with two roots that are functions of t. We have to pick the one which has no singularity at $t = 0$. When $s_1 = 0$ we have a linear equation which may have a single root. When $s_1 = 0$ we have the equation $(1 - s_0 t)x = t$ so the single root is

$$\sigma = \frac{t}{1 - s_0 t}$$

When $s_1 = 1$ we have the equation $tx^2 - (1 - s_0 t)x + t = 0$ with roots

$$\sigma = \frac{1 - s_0 t - \sqrt{(1 - s_0 t)^2 - 4t^2}}{2t}, \quad \sigma^* = \frac{1 - s_0 t + \sqrt{(1 - s_0 t)^2 - 4t^2}}{2t}$$

The two roots multiply to 1 so $\overline{\sigma} = \sigma^*$. Later we shall have a little table tabulating all the results. Note that in the relationship between the labelled and unlabelled q_0 we have

$$q_0 = \frac{t}{at - b\sigma} \widehat{q_0}$$

where σ is a function containing a square root, i.e. not a polynomial. However we can go at the problem from a different angle to obtain a polynomial $f(a, t)$ for which $B(fq_0) = c$ which is not quite zero but it's a simpler relationship. This will also illustrate the group approach together with the kernel method. The two cases that we need to work on both have $s_{-1} = s_1 = 1$ with s_0 being 0 or 1. This means that s is left invariant under sending $x \mapsto \overline{x}$.

2.2.3

In this section we were able to find explicit solutions for q_0, q and also B-operator relationships for q and $\widehat{q}$. In higher dimensions this may not be quite possible but we should still be able to find B-operator relationships for the equivalent of q_0 and $\widehat{q_0}$.

3 Deriving the functional equation for the 2D unlabelled random walks

We begin constructing a functional equation for Q, the generating function for the quarter-plane walks with steps from S (we'll use it for both the step set and the laurent polynomial). We use the inclusion-exclusion principle. We start with 1 (I'm not sure exactly where it comes). When we are far from the origin and the axes we can step in any of the directions from S. So we'll add tSQ because the walks will now have 1 more step and any of the walks counted by Q should be in there as well.

However, we've overdone it because the expression tSQ also counts those walks that end up below the x_1 axis or on the left of the x_2 axis. We must subtract those invalid walks. A walk will become invalid with respect to the x_1 axis if it start at coordinate $(i, 0)$ for some i, that is the x_1 axis and the next step leads it below the x_1 axis, that is with negative x_2 coordinate. The walks starting on the x_1 axis are those counted by Q_2^0. The steps in the negative x_2 directions are those counted by S_2^{-1}. The term we need to remove is $t\overline{x_2}S_2^{-1}Q_2^0$

where the t is due to the increase in step number. Analogously we need to remove the term $\overline{x_1}S_1^{-1}Q_1^0$.

However, there is a problem. We've removed too many terms, namely those that count walks on the origin and go in the SW direction, because this is the common direction counted by both the terms in the previous paragraph. Thus we need to add in the term involving walks starting at the origin (counted by $Q_{1,2}^{0,0}$) and moving in the SW direction (steps counted by $S_{1,2}^{-1,-1}$). The term is $t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}$. Since this covers all cases we immediately have the functional equation

$$Q = 1 + tSQ - t\overline{x_2}S_2^{-1}Q_2^0 - t\overline{x_1}S_1^{-1}Q_1^0 + t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}$$

4 Deriving the functional equation for the 2D labelled random walks

To derive the functional equation for the labelled random walks we start with the equation for the unlabelled walks and modify it to add terms counting interactions with the walls. Each of the terms will rely on inclusion-exclusion to arrive at a valid expression. First, we need to identify which terms will count walks that will have to be labelled on their step, remove those terms and add them back in with a multiplication by a_i.

We start with identifying which terms will be labelled on the next step due to an interaction with the x_1-axis. We have two cases

- Walks ending on the x_1-axis (counted by Q_2^0) moving in the positive or negative x_1 direction (steps counted by S_2^0). The term is $tS_2^0Q_2^0$.
- Walks ending one unit above the x_1-axis (counted by Q_2^1) moving in the negative x_2 direction (steps counted by S_2^{-1}). The term is $tS_2^{-1}Q_2^1$.

However, we've over-reached in this tally and we need to remove those terms that count invalid quarter-plane walks, namely those that also end up beyond the x_2 axis on the next step.

- Walks ending on the x_1-axis and the x_2-axis (counted by $Q_{1,2}^{0,0}$) travelling in the negative x_1 direction (steps counted by $S_{1,2}^{-1,0}$). The term is $t\overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0}$
- Walks ending one unit above the x_1-axis and on the x_2-axis (counted by $Q_{1,2}^{0,1}$) traveling in the negative x_2 direction and negative x_1 direction. The term is $t\overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1}$.

So the quarter-plane walks that need to be labelled with a_2 for every time they touch the x_1 axis are given by the term

$$t(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0})$$

We need to remove this term and add it back again with a label a_2 because the term counts those walks that will interact with the x_1 axis on their next step. Thus overall we need to add

$$t(a_2 - 1)(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0})$$

to the unlabelled functional equation. Analogously we need to add

$$t(a_1-1)(S_1^{-1}Q_1^1+S_1^0Q_1^0-\overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0}-\overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0})$$

to the equation. However, we've mislabelled walks that would end at the origin on their next step. We've given them a_1 as a label in one case and a_2 as a label in the other case. Furthermore we've removed two copies of each due to the two terms. Therefore, we need to add back one of the terms, remove the terms with labels a_1 and a_2 and relabel them properly with label a_1a_2. There are three types of walks ending on the origin

- Walks ending on point $(0,1)$ (counted by $Q_{1,2}^{0,1}$) and then stepping South, i.e. step is $(0,-1)$ (counted by $S_{1,2}^{0,-1}$). The term is $tS_{1,2}^{0,-1}Q_{1,2}^{0,1}$.
- Walks ending on point $(1,0)$ (counted by $Q_{1,2}^{1,0}$) and then stepping West, i.e. step is $(-1,0)$ (counted by $S_{1,2}^{-1,0}$). The term is $tS_{1,2}^{-1,0}Q_{1,2}^{1,0}$.
- Walks ending on point $(1,1)$ (counted by $Q_{1,2}^{1,1}$) and then stepping Southwest, i.e. step is $(-1,-1)$ (counted by $S_{1,2}^{-1,-1}$). The term is $tS_{1,2}^{-1,-1}Q_{1,2}^{1,1}$.

We'll need to multiply the terms by the following

$$a_1a_2-a_1-a_2+1=(a_1-1)(a_2-1)$$

in order to take care of the relabelling. Finally the term that we need to add to the equation is

$$t(a_1-1)(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1}+S_{1,2}^{-1,0}Q_{1,2}^{1,0}+S_{1,2}^{-1,-1}Q_{1,2}^{1,1})$$

This is our final step in the construction becuase we've taken care of all possible cases. The argument will be generalized to d dimensions in a later section. The lengthy inclusion-exclusion-type derivation yields the labelled equation

$$\begin{aligned}Q=1&+tSQ-t\overline{x_2}S_2^{-1}Q_2^0-t\overline{x_1}S_1^{-1}Q_1^0+t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}\\&+t(a_1-1)(S_1^{-1}Q_1^1+S_1^0Q_1^0-\overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0}-\overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0})\\&+t(a_2-1)(S_2^{-1}Q_2^1+S_2^0Q_2^0-\overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1}-\overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0})\\&+t(a_1-1)(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1}+S_{1,2}^{-1,0}Q_{1,2}^{1,0}+S_{1,2}^{-1,-1}Q_{1,2}^{1,1})\end{aligned}$$

5 Simplifying the 2D functional equation

As we saw in the previous section the functional equation for labelled 2D walks is

$$\begin{aligned}Q=1&+tSQ-t\overline{x_2}S_2^{-1}Q_2^0-t\overline{x_1}S_1^{-1}Q_1^0+t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}\\&+t(a_1-1)(S_1^{-1}Q_1^1+S_1^0Q_1^0-\overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0}-\overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0})\\&+t(a_2-1)(S_2^{-1}Q_2^1+S_2^0Q_2^0-\overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1}-\overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0})\\&+t(a_1-1)(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1}+S_{1,2}^{-1,0}Q_{1,2}^{1,0}+S_{1,2}^{-1,-1}Q_{1,2}^{1,1})\end{aligned}$$

The equation can be greatly simplified by eliminating terms that involve quarter-plane walks not ending on the axes or the origin. For that we'll first extract some coefficients.

$$\begin{aligned}
Q_1^0 &= 1 + ta_1(S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x}_2 S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0}) \\
&\quad + ta_1(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}) \\
Q_2^0 &= 1 + ta_2(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x}_1 S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0}) \\
&\quad + ta_2(a_1-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}) \\
Q_{1,2}^{0,0} &= 1 + ta_1a_2(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})
\end{aligned}$$

There is a hidden pattern in the form of the sumands in these extractions. Let

$$A = SQ - \overline{x_2}S_2^{-1}Q_2^0 - \overline{x_1}S_1^{-1}Q_1^0 + \overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}$$

Extracting coefficients yields

$$\begin{aligned}
A_1^0 &= S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0} \\
A_2^0 &= S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0} \\
A_{1,2}^{0,0} &= S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}
\end{aligned}$$

A quick visual inspection shows that the A terms are involved in the functional equation and the Q-extractions. This allow us to re-write the above equations in a much neater format. Using $b_i = a_1 - 1$ the equations become

$$\begin{aligned}
Q &= 1 + tA + tb_1A_1^0 + tb_2A_2^0 + tb_1b_2A_{1,2}^{0,0} \\
Q_1^0 &= 1 + ta_1A_1^0 + ta_1b_2A_{1,2}^{0,0} \\
Q_2^0 &= 1 + ta_2A_2^0 + ta_2b_1A_{1,2}^{0,0} \\
Q_{1,2}^{0,0} &= 1 + ta_1a_2A_{1,2}^{0,0}
\end{aligned}$$

Assuming the A-terms are non-zero we can back-substitute the A's in order to obtain an equation relating Q, Q_1^0, Q_2^0 and $Q_{1,2}^{0,0}$. In other words, we'll obtain a functional equation that relates Q to Q's counting walks ending on the axes or the origit. First

$$A_{1,2}^{0,0} = \frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}$$

and then

$$\begin{aligned}
A_1^0 &= \frac{Q_1^0 - 1}{ta_1} - (a_2-1)A_{1,2}^{0,0} = \frac{Q_1^0 - 1}{ta_1} - (a_2-1)\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2} \\
A_2^0 &= \frac{Q_2^0 - 1}{ta_2} - (a_1-1)A_{1,2}^{0,0} = \frac{Q_2^0 - 1}{ta_2} - (a_1-1)\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}
\end{aligned}$$

We can substitute these expressions into the initial equation along with $c_i = b_i/a_i = 1 - \overline{a_i}$

$$\begin{aligned} Q &= 1 + tA + tb_1A_1^0 + tb_2A_2^0 + tb_1b_2A_{1,2}^{0,0} \\ &= 1 + tA + tb_1\left(\frac{Q_1^0 - 1}{ta_1} - b_2\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}\right) + tb_2\left(\frac{Q_2^0 - 1}{ta_2} - b_1\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}\right) + tb_1b_2\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2} \\ &= 1 + tA + c_1(Q_1^0 - 1) + c_2(Q_2^0 - 1) - c_1c_2(Q_{1,2}^{0,0} - 1) \end{aligned}$$

Expanding the definition of A and re-arranging terms gives the following functional equation for Q relating it to Q_1^0, Q_2^0 and $Q_{1,2}^{0,0}$.

$$Q(1 - tS) - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2}$$

The form is suggestive so for $I \subseteq [2]$ we let

$$K_I = c_I - t\overline{x_I}S_I^{-\mathbb{I}}$$

be the kernels with respect to S. When $I = \varnothing$ we just write $K = 1 - tS$. This allows us to re-write the above equation in the following compact way

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \overline{a_1a_2}$$

This equation is an important milestone in further analysing the labelled quarter plane walks. Before we go on though, we need to say a few words about the cases when one or more of the A's is 0. If $A_1^0 = 0$ or $A_2^0 = 0$, this also means $A_{1,2}^{0,0} = 0$. This in turn makes the respective Q's equal to 1. For example, if only $A_{1,2}^{0,0} = 0$ then $S_{1,2}^{-1,-1}$, $S_{1,2}^{-1,0}$ and $S_{1,2}^{0,-1}$ are all zero, together with $Q_{1,2}^{0,0} = 1$.(Note: I now realize that this is not completely necessary. One could tell aprior if one or more of the Q's will be zero based on the step sets we have. We either have no positive steps or no negative steps for $A_{1,2}^{0,0}$ to be identially zero. This also suggest that we might be able to perform a different derivation that will focus on going forward rather than backward like this). Then

$$A_1^0 = \frac{Q_1^0 - 1}{ta_1}, \quad A_2^0 = \frac{Q_2^0 - 1}{ta_2}$$

So we obtain

$$\begin{aligned} Q &= 1 + tA + tb_1A_1^0 + tb_2A_2^0 + tb_1b_2A_{1,2}^{0,0} \\ &= 1 + tA + tb_1\frac{Q_1^0 - 1}{ta_1} + tb_2\frac{Q_2^0 - 1}{ta_2} \\ &= 1 + tA + c_1(Q_1^0 - 1) + c_2(Q_2^0 - 1) \end{aligned}$$

Expanding the definition of A and re-arranging terms gives the following functional equation for Q relating it to Q_1^0 and Q_2^0.

$$Q(1 - tS) - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 = \overline{a_1a_2} - c_1c_2$$

We have a similar situation if any of the other A's are zero.

6 Generalization to d dimensions.

This section needs a major re-work since I'm not 100 percent clear on the validity of this derivation.

The notation so far has helped push the derivation of the functional equation forward. It also gives us a good idea how to generalize the building process for the functional equation to more dimensions and more general step sets. From now on we fix the numbe of dimensions to d and the step lengts to 1 in any of the coordinates. The 2D derivation used progressive coefficient extractions to signify interactions with more and more axes simultaneously. In d dimensions we have interactions with hyperplanes defined by setting a single coordinate to zero. i.e. the hyperplanes defined by x_i for i between 1 and d. We'll first derive the unlabelled version of the functional equation. We start with Q on the LHS and 1 on the RHS and progressively add/subtract terms to/from the RHS.

First we add the term extending arbitrary p-ve orthant walks (counted by Q) by one step in any of the directions (counted by a Laurent polynomial S). This gives a term tSQ but also intorduces invalid walks that reach beyond the boundary of the p-ve orthant. That is one or more of their coordinates become negative. We need to remove those terms. For coordinate $1 \leqslant i \leqslant d$ we remove those walks that step beyond the hyperplane $x_i = 0$. These are all the walks ending with coordinate $x_i = 0$ (counted by Q_i^0) stepping in the negative x_i-direction (steps counted by S_i^{-1}). The term that we need to add to the equation so far is

$$\sum_{i=1}^{d}(-1)^1\overline{x_i}S_i^{-1}Q_i^0$$

We've removed too many walks though. To be specific we've over-removed the walks that end at points with at leats 2 coordinates equal to zero. This will be the basis step in the following induction. Assume that at step $1 \leqslant k \leqslant d$ in the process we add the term

$$\sum_{\substack{J\subseteq[d]\\|J|=k}}(-1)^{|J|}\overline{x_J}S_J^{-\mathbb{I}}Q_J^{\mathbb{O}}$$

where we've over-compensated for the walks that end on points with at least $k+1$ coordinates equal to 0. Let J be a set of $k+1$ coordinates. We'll determine the compensating term for those walks that end at points with coordinates $x_j = 0$, $j \in J$ (some more coordinates may be zero too). These are counted by the term $Q_J^{\mathbb{O}}$. We need to compensate for steps counted in the previous step, that is, steps that go in the negative x_j direction for each $j \in J$ (steps counted by $S_J^{-\mathbb{I}}$). The term is $tS_J^{-\mathbb{I}}Q_J^{\mathbb{O}}$. Adding all these terms up yields the compensatory term

$$\sum_{\substack{J\subseteq[d]\\|J|=k+1}}(-1)^{|J|}\overline{x_J}S_J^{-\mathbb{I}}Q_J^{\mathbb{O}}$$

where we've again over-compensated for those coordinates that end at points with at least $k+2$ coordinates, unless this is larger than d. This completes the induction. Combining all the compesatory terms into one major term gives

$$A = \sum_{J\subseteq[d]}(-1)^{|J|}\overline{x_J}S_J^{-\mathbb{I}}Q_J^{\mathbb{O}}$$

and the unlabelled problem has an equation that reads as

$$Q = 1 + tA. \tag{1}$$

We can also prove this combinatorially. All terms in A except the first term SQ deal with removing walks that end at hyperplanes and intersections of hyperplanes. Suppose we have an orthant walk w ending at a point with exactly k coordinates equal to zero. Let I be list those coordinates. All the other coordinates are non-zero, so we are allowed to step in those negative directions. Thus to make an invalid walk out of w, we need to step in at least one negative x_i direction for some $i \in I$. This type of walk is counted on the RHS in every term $S_J^{-\mathbb{I}} Q_J^{\mathbb{O}}$ where $J \cap I \neq \varnothing$. There are $|I|$ choose $|J| = k$ yada yada yada. They all add up and we get 1.

I wonder if there is a direct way to get this.

We first start with walks ending on no hyperplanes. Then we remove those that end on a hyperplane and go strictly beyond on the next step. Then we add back those that end on two hyperplanes simultaneously and go strictly beyond both. This means they have to go in direction that is negative with respect to both. Suppose that we've compensated for walks up to but not including those that end on k hyperplanes. We've overcompensated for those that end on k hyperplanes and go strictly beyond all k of them. This means that we need to compensate back for those walks that end on k hyperplanes and take a step in the negative direction with respect to all k hyperplanes. The term describing walks that end on k hyperplanes (listed in I) and go in the negative direction with respect to all k of them is $tS_I^{-\mathbb{I}} Q_I^{\mathbb{O}}$. The inclusion-exclusion process yields the formula directly.

Next we start working on the labelled version. What does A represent? Here A stands for allowed walks in the functional equation for the generating function. Indeed A contains only valid walks. Because $Q = 1 + tA$ this means that A contains all allowed walks except the empty walk. The way we build the next part is to consider which allowed walks we are going to label. First we label with a_i those that end on hyperplane $x_i = 0$ and move in an allowed direction. Those walks are counted by A_i^0. We need to multiply this term by b_i to remove the unlabelled ones and add in the labelled ones. However, we've mislabelled those allowed walks that end on two hyperplanes. Those are counted by $A_{i,j}^{0,0}$. They've been labelled by a_i, a_j and removed twice from the allowed walks, so we need to multiply the term by $b_i b_j$. Suppose that we've completed the process up to but not including step k. We've overcompensated for allowed walks on k hyperplanes listed in I. Those are counted by $A_I^{\mathbb{O}}$. We need to multiply this by some term involving labels to offset the wrong labels from the previous step.

Because the relabelling in the previous step for a given term $A_J^{\mathbb{O}}$ ends up with label a_J, the previous step has mislabelled the walks counted by $A_I^{\mathbb{O}}$ with labels a_J for $J \subseteq I$ and $|J| < |I|$. That is we need to multiply $A_I^{\mathbb{O}}$ with

$$a_I - \sum_{J \subseteq I, |J| < |I|} (-1)^{|I|-|J|} a_J = b_I$$

This gives the labelled equation

$$Q = 1 + t \sum_{I \subseteq [d]} b_I A_I^{\mathbb{O}} \tag{2}$$

We can write out the coefficient extraction explicitly as

$$Q = 1 + t \sum_{\substack{I,J\subseteq[d]\\ I\cap J=\varnothing}} b_I \overline{x_J} \sum_{K\subseteq I} \left(S^{-\mathbb{I},-\mathbb{I}}_{J,K} Q^{\mathbb{O},-\mathbb{I}}_{J,K} + S^{-\mathbb{I},\mathbb{O}}_{J,K} Q^{\mathbb{O},\mathbb{O}}_{J,K} \right)$$

although this doesn't seem to be necessary. Next, we perform coefficient extractions on equation (2) to obtain the following formula

$$Q^{\mathbb{O}}_J = 1 + t \sum_{I\subseteq[d]} b_I A^{\mathbb{O}}_{I\cup J} \tag{3}$$

$$= 1 + t a_J \sum_{I\subseteq[d]\backslash J} b_I A^{\mathbb{O},\mathbb{O}}_{I,J} \tag{4}$$

even though the second equation takes a little bit of work. Before we continue we have to make sure we know what to do if any of the A's are equal to zero. The following lemma characterizes this situation

Lemma 1. $A^{\mathbb{O}}_I \equiv 0$ if and only if $S^{-\mathbb{I},P}_{J,I} \equiv 0$ for all $J \subseteq [d]\backslash I$ and $P \in \{-1, 0\}^{|I|}$.

alternatively, since $S^{-\mathbb{I},P}_{J_1,I} \equiv 0 \Rightarrow S^{-\mathbb{I},P}_{J_2,I} \equiv 0$ for $J_1 \subseteq J_2$ based on simple extraction, then it only suffices to look at S^P_I Therefore we can re-write the lemma as

Lemma 2. $A^{\mathbb{O}}_I \equiv 0$ if and only if $S^P_I \equiv 0$ for all $P \in \{-1, 0\}^{|I|}$. In particular $A^{\mathbb{O}}_I \equiv 0 \Rightarrow S^{-\mathbb{I}}_I \equiv 0$.

The second part of the lemma guarantees that if an A is zero then the corresponding S will be zero in the specific term K in the master equation. The next part takes some effort but we can use backward induction on $|I|$ to show that

$$A^{\mathbb{O}}_I = \frac{(-1)^{|I|}}{t b_I} \sum_{I\subseteq J\subseteq[d]} (-1)^{|J|} c_J (Q^{\mathbb{O}}_J - 1) \tag{5}$$

Substituting these terms back into equation (2) yields the following calculation

$$Q = 1 + tA + t \sum_{\varnothing\neq I\subseteq[d]} b_I \frac{(-1)^{|I|}}{t b_I} \sum_{I\subseteq J\subseteq[d]} (-1)^{|J|} c_J (Q^{\mathbb{O}}_J - 1) \tag{6}$$

$$= 1 + tA + \sum_{\varnothing\neq I\subseteq[d]} (-1)^{|I|} \sum_{I\subseteq J\subseteq[d]} (-1)^{|J|} c_J (Q^{\mathbb{O}}_J - 1) \tag{7}$$

For each I we are taking all sets inbetween I and $[d]$ for the latter sum. How many times do we see every subset J? We see it once for every subset I. There are $2^{|J|} - 1$ such subsets because $I \neq \varnothing$. Since J is non-empty, the number is odd so we have

$$Q = 1 + tA + \sum_{\varnothing\neq J\subseteq[d]} (-1)^{|J|+1} c_J (Q^{\mathbb{O}}_J - 1) \tag{8}$$

$$= 1 + t \sum_{J\subseteq[d]} (-1)^{|J|} \overline{x_J} S^{-\mathbb{I}}_J Q^{\mathbb{O}}_J + \sum_{\varnothing\neq J\subseteq[d]} (-1)^{|J|+1} c_J (Q^{\mathbb{O}}_J - 1) \tag{9}$$

$$= 1 + \sum_{\varnothing\neq J\subseteq[d]} (-1)^{|J|} c_J + tSQ - \sum_{\varnothing\neq J\subseteq[d]} (-1)^{|J|} (c_J - t\overline{x_J} S^{-\mathbb{I}}_J) Q^{\mathbb{O}}_J \tag{10}$$

The sum of c_J's combines with 1 to create $\overline{a_{[d]}}$. Rerranging everything else to the LHS leaves us with the d-dimensional master equation

Theorem 3. *Given a step set in $d \geqslant 1$ dimensions with Laurent polynomial S, the labelled generating function satisfies the following functional equation*

$$\sum_{I \subseteq [d]} (-1)^{|I|}(c_I - t\overline{x_I}S_I^{-\mathbb{I}})Q_I^{\mathbb{O}} = \overline{a_{[d]}}$$

or if we let $K_I = c_I - t\overline{x_I}S_I^{-\mathbb{I}}$ we have

$$\sum_{I \subseteq [d]} (-1)^{|I|}K_I Q_I^{\mathbb{O}} = \overline{a_{[d]}}.$$

We also have a corollary that takes care of all the cases where $S_I^{-\mathbb{I}} = 0$. We have a modified formula

Corollary 4. For everything as in the theorem we have

$$\sum_{\substack{I \subseteq [d] \\ A_I^{\mathbb{O}} \neq 0}} (-1)^{|I|}(c_I - t\overline{x_I}S_I^{-\mathbb{I}})Q_I^{\mathbb{O}} = f(a_1, a_2, \ldots, a_d)$$

where

$$f(a_1, a_2, \ldots, a_d) = \overline{a_{[d]}} - \sum_{\substack{I \subseteq [d] \\ A_I^{\mathbb{O}} \neq 0}} (-1)^{|I|}c_I$$

which could easily be simplified further, now that I think about it.

7 Using Operator Notation to Simplify the Derivation

There is a better way for obtaining the functional equation and the key is to think in terms of operators that we apply on generating functions. If Q is the generating function for all the quarter plane walks with step set S, then for the Laurent polynomial S for the steps, tSQ is the generating function for all quarter plane walks extended by each of the steps in S counting with t the fact that they are all now one step longer. Instead of going through an inclusion exclusiong principle instead we can do it step by step. Let

$$E_i = 1 - \overline{x}_i[\overline{x}_i]$$

be the extraction operator with respect to x_i where 1 is interpreted as the identity operator. Then $E_1(tSQ)$ takes all the walks in tSQ and removes those with endpoints that have a negative x_1 coordinate (in this case -1). After the operator is applied all the walks remaining in the expression a valid quarter plane walks with respect to the x_1 coordinate. Then we can apply E_2, that is calculate $E_2E_1(tSQ)$. By induction we see that $E_d \ldots E_1(tSQ)$ counts all the valid quarter plane walks except the trivial one whose term is just 1. For the unlabelled quarter plane walks in d dimensions we can write

$$Q = 1 + E_d \ldots E_1(tSQ) = 1 + tE(SQ)$$

where we let $E = E_d \dots E_1$ Next, we can take care of labelling. Since $tE(SQ)$ is the generating function for all the quarter plane walks except the trivial one, we can see this is as a process where we take a walk in Q and append a step from S, while E clears the invalid walks. To do the labelling we need to take the walks that have arrived at an axis and label them. For this we define another operator. Let

$$L_i = 1 + b_i[x_i^0]$$

denote the labelling operator. The way it works is by applying it on $tE(SQ)$. This takes the valid quarter-plane walks as one summand. The other summand is taking those walks arriving at the $x_i = 0$ hyperplane and labelling them by removing them and adding them back in with the label a_i. That's why we have b_i there. To label the walks for all the axes we just apply the L_i operators sequentially. L_1 takes care of the $x_1 = 0$ hyperplane labelling. Then L_2 takes the result and takes care of the $x_2 = 0$ hyperplane labelling. Continuing on by induction we have that

$$Q = 1 + tLE(SQ)$$

where $L = L_d \dots L_1$. The beauty of thinking in terms of operators is that we immediately think about inverses. This will be covered more extensively in a later section. For now it suffices to say that the E_i operators are not invertible because

$$E_i c\overline{x}_i = c\overline{x_i} - \overline{x}_i[\overline{x}_i]c\overline{x}_i = 0$$

and that the L_i's are invertible with inverse

$$L_i^{-1} = M_i = 1 - c_i[x_i^0].$$

To see this we just calculate

$$\begin{aligned} M_i L_i &= (1 - c_i[x_i^0])(1 + b_i[x_i^0]) = 1 - c_i[x_i^0] + b_i[x_i^0] - c_i[x_i^0]b_i[x_i^0] \\ &= 1 + (b_i - c_i - c_i b_i)[x_i^0] = 1 + (b_i - c_i a_i)[x_i^0] \\ &= 1. \end{aligned}$$

Also it's important to mention that the L_i operators are commutative amongst each other and similarly with M_i, and with E_i. Also, each operator is commutative with any other one that doesn't pertain to the same variable. We see that

$$L^{-1} = (L_d \dots L_1)^{-1} = M_d \dots M_1 = M$$

We can now apply this to the functional equation for Q to get

$$MQ = M(1) + tE(SQ)$$

which we can write as

$$(M - tES)Q = M(1) = \overline{a}_{[d]}$$

In the case of steps of size 1 and labelling the hyperplane contacts we have

$$MQ = (1 - c_1[x_1^0]) \dots (1 - c_d[x_d^0])Q = \sum_{I \subseteq [d]} (-1)^{|I|} c_I [x_I^{\mathbb{0}}]Q = \sum_{I \subseteq [d]} (-1)^{|I|} c_I Q_I^{\mathbb{0}}$$

and

$$\begin{aligned} ESQ &= (1-\overline{x}_1)[\overline{x}_1])\dots(1-\overline{x}_d)[\overline{x}_d 1])(SQ) \\ &= \sum_{I\subseteq[d]} (-1)^{|I|}\overline{x}_I[x_I^{-\mathbb{I}}](SQ) \\ &= \sum_{I\subseteq[d]} (-1)^{|I|}\overline{x}_I S_I^{-\mathbb{I}} Q_I^{\mathbb{O}} \end{aligned}$$

This calculation finally allows us to write the functional equation in the kernel form that we had before.

8 Generalizations are Handled Easily

Using the operator notation we can handle generalizations quite easily. First we can consider a more general step set. Let n_i be non-negative integers for $1 \leqslant i \leqslant d$. Then let S be a step set for which no step backtracks more than n_i units in the negative x_i direction. The positive direction is bounded. Then we can write the extraction operators as

$$E_i = 1 - \overline{x}_i[\overline{x}_i] - \dots - \overline{x}_i^{n_i}[\overline{x}_i^{n_i}]$$

The reason why this works is that this operator must remove all walks that are not valid with respect to the x_i coordinate. Starting at the $x_i = 0$ hyperplane, any walks can step at most a distance n_i away from it in the x_i coordinate. That's why we need to remove all those terms. An example of this process could be the upper half-plane walks with $E_1 = 1$ and $E_2 = 1 - \overline{x}_2[\overline{x}_2]$ and unrestricted quarter-plane walks have $E_1 = E_2 = 1$. Of course we could easily generalize to exclude certain step sizes. For example, if the walks have a certain step length, they might be able to jump over wide boundaries (this might be useful in some models). Another example is where we might want to look at all quarter plane walks that don't step at a certain point with coordinates (m, n). Then we can write an operator

$$E(m,n) = 1 - x_1^m x_2^n [x_1^m x_2^n]$$

that we'd apply on SQ to remove those walks that end on (m, n).

Next we take care of labelling. We might want to label the interactions not just with the boundaries but also with specific hyperplanes. For example, if we want to label the boundary interactions and also the interactions with the closest 2 hyperplanes near the $x_i = 0$ hyperplane in the direction of the x_i axis we'd write

$$L_i = 1 + b_{i,0}[x_i^0] + b_{i,1}x_i^1[x_i^1] + b_{i,2}x_i^2[x_i^2]$$

Of course we can also easily label the interactions with specific points. If we want to label the interactions with point (m, n) with a label $a_{m,n}$ we write

$$L(m,n) = 1 + b_{m,n}x_1^m x_2^n [x_1^m x_2^n]$$

It might be interesting to consider the interactions of quarter plane walks with the line $x_2 = (q/r)x_1$ for some coprime integers q, r. For this we need the label

$$L = 1 + b\sum_{k=0}^{\infty} x_1^{qk} x_2^{rk} [x_1^{qk} x_2^{rk}]$$

This is going to get a little crazy right away so I'm sorry. I just thought of the most general thing I could. All these operators can be thought of as belonging to a family of more generalized operators

$$\Gamma_k(x_1, \ldots, x_d) = \sum_{\substack{I \subseteq [d] \\ |I|=k}} \sum_{J \in \mathbb{Z}^k} \gamma_{I,J} x_I^J [x_I^J]$$

For example the last one can be written as

$$L + 1 + b\Gamma_1(x_1^q x_2^r)$$

for $\gamma_k = 1$ and the first one in the section as

$$E_i = 1 - \Gamma_1(\overline{x}_1)$$

with the sequence $\gamma_k = 1$ for $0 \leqslant k \leqslant n_i$. It's more of a thought right now, not something I've really looked into.

9 Properties of the operators

In this section we determine properties of some of the operators like their null spaces, eigenvalues and eigenvectors. We'll assume we are dealing with steps of size 1 and labeling the interactions with the hyperplanes $x_i = 0$. First we take $E = 1 - \overline{x}[\overline{x}]$. We have

$$0 = E(f) = f - \overline{x} f_{-1} = 0$$

which means that f is in the null space of E if $f = c\overline{x}$ for some constant. This shows that E itself is not invertible. What about ES? We have

$$S = S_{-1}\overline{x} + S_0 + S_1 x$$

for some S_{-1}, S_0, S_1 constant with respect to x and not all zero. Then

$$0 = ES(f) = Sf - \overline{x}(S_{-1} f_0 + S_0 f_{-1} + S_1 f_{-2})$$

This implies that

$$f = \frac{(S_{-1} f_0 + S_0 f_{-1} + S_1 f_{-2})}{xS}$$

where the numerator is a constant and the denominator is a polynomial (at most a quadratic), both with respect to x. This means that f is a power series because we can expand by partial fractions and expand using the general binomial theorem. Therefroe $f_{-1} = f_{-2} = 0$ and

$$f = \frac{cS_{-1}}{xS}$$

This gives two cases. In the first case $S_{-1} = 0$ and the null-space of ES is 0, so the operator is invertible. It's equivalent to the identity on power series f but it gives

$$ES(f) = Sf - \overline{x}(S_0 f_{-1} + S_1 f_{-2})$$

on Laurent series. In the second case $S_{-1} \neq 0$ and $f = c/(xS)$ is in the null-space of ES, so ES is not invertible.

Next we look at eigenvalues. Let λ be constant with respect to x. Then we have to solve the equation

$$ES(f) = Sf - \overline{x}(S_{-1} f_0 + S_0 f_{-1} + S_1 f_{-2}) = Sf - c\overline{x} = \lambda f$$

Therefore

$$f = \frac{c}{x(S - \lambda)}.$$

The numerator is a constant, the denominator is a polynomial in x and therefore f is defined by a power series in x, so $f_{-1} = f_{-2} = 0$ and therefore the eigenfunctions are given by

$$f = \frac{cS_{-1}}{x(S - \lambda)}$$

for every λ. If $S_{-1} = 0$ then there are no eigenvalues and eigenfunctions. If $S_{-1} \neq 0$ then we have an eigenfunction for every λ. I guess the operator has a continuous spectrum then?

It is only natural to consider the properties of the operator $ES = E_1 \dots E_d S$. I think here one would input $f = g/S$ for some other function g. Then we have

$$ES(f) = ESg/S = Eg$$

so we only need to consider the properties of E itself. Certainly the null-space of E will contain the linear combination of null-spaces for E_i for if $E_i g = 0$, then $Eg = 0$. That is, $g = g'/x_i$ for some g', not dependent on x_i. This means that the products of $\overline{x}_i$'s will form part of the null-space for E. If g is a linear combination of terms of that form, $Eg = 0$. Therefore let $g = g_1 + g_2$ where g_1 is a multivariate series where $[\overline{x}_i]g_1 = 0$ for every i and g_2 is a linear combination of monomials composed of $\overline{x}_i$'s. Then clearly

$$Eg = Eg_1 + Eg_2 = Eg_1$$

and since $[\overline{x}_i]g_1 = 0$ for every i, $Eg_1 = g_1$. Thus we conclude that if g is in the null-space of E it has to be a linear combination of monomials composed of nonrepeating x_i's. Thus we find that

$$\text{Null}(ES) = \left\{ \frac{m}{S} \mid m \text{ is a monomial composed of non-repeating } x_i\text{'s} \right\}$$

We have a similar situation for the eigenpairs. For each λ, $f/(S - \lambda)$ is an eigenfunction for each $f \in \text{Null}(ES)$.

Finally we should look at the invertibility and other properties of the operator $1 - tLES$ or even $M - tES$.

10 Using the formal $\mathcal{L}^{-1}$ operator for iteration

In a previous section we had the equation

$$(1 - tLES)Q = 1$$

Let $D = LE$. Then we can think of the formal inverse of this operator against some norm $||tDS||$ or even formally. Formally

$$(1 - tDS)^{-1} = \frac{1}{1 - tD} = \sum_{k=0}^{\infty} t^k (DS)^k$$

so we can write the solution for Q as

$$Q = \sum_{k=0}^{\infty} t^k (DS)^k (1)$$

where $(DS)^k(1) = DS(\ldots(DS(1))\ldots)$. This allows us to compute the series expansion for Q directly using an iteration. The next coefficient is the previous one with DS applied to it. It can be done fairly efficiently on a computer. Interestingly enough this is just a formalization of the brute force method of determining the functional equation - by taking a previous set of walks, extending them by 1 step, removing the ones that fall outside of the quarter plane and finally labelling the ones that have interacted with the hyperplanes $x_i = 0$ in the process. We could attempt to write a closed form for the operator application but it becomes messy fairly fast, even for one dimension.

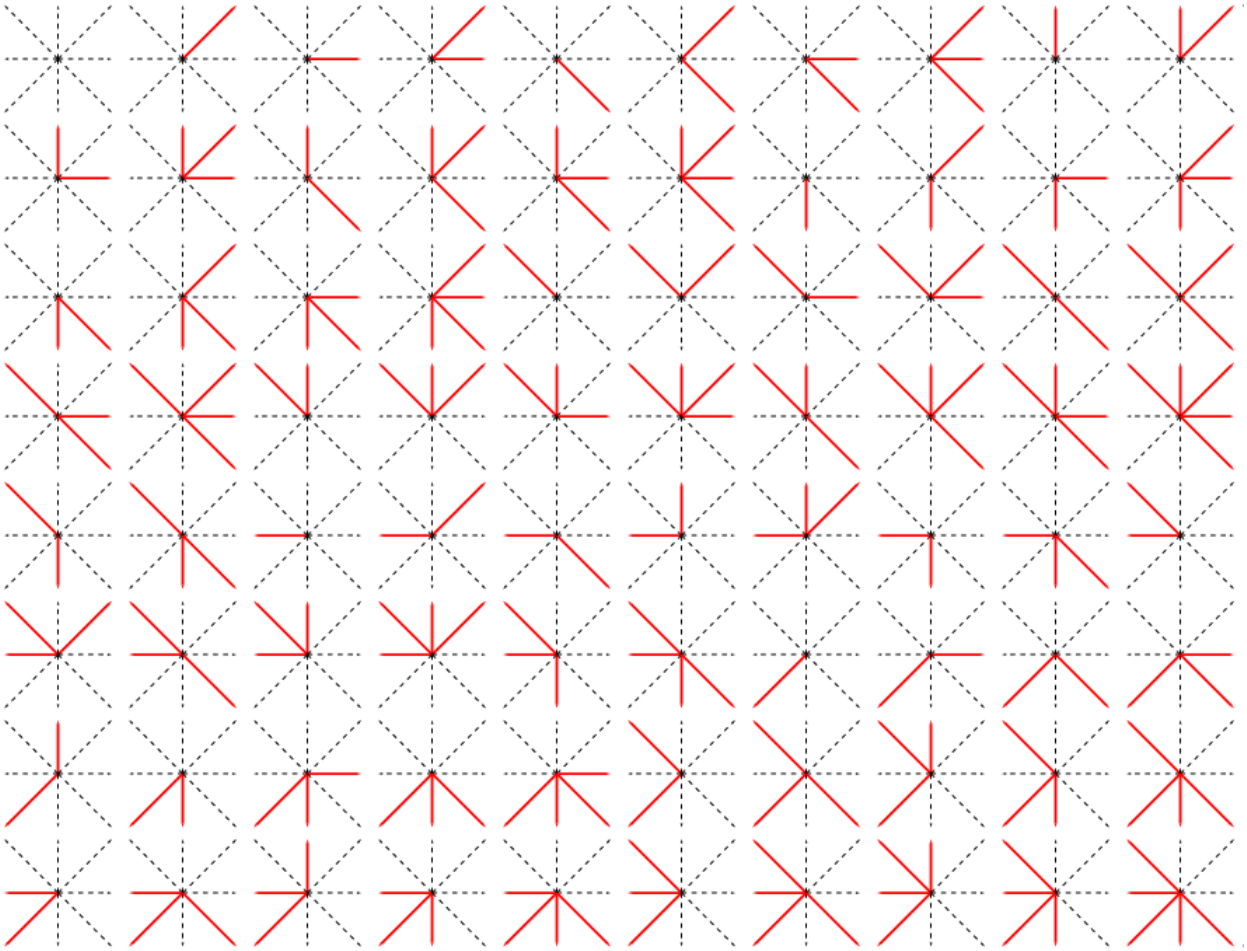
11 Investigating relationships between G, G_1, G_2 and $G_{1,2}$

In this work we'll focus on investigating the properties of the function $G(a_1, a_2) = Q^{0,0}_{1,2}(a_1, a_2)$. Let $G_1 = G(1, a_2)$, $G_2 = G(a_1, 1)$ and $G_{1,2} = G(1, 1)$. We could also see these as coefficient extractions of G with respect to b_1, b_2 and $b_1 b_2$, i.e. $G_{1,2} = [b_1^0 b_2^0]G$. In previous works it has been shown that there is a relationship between these quantities depending on the particular step set S. In some cases (like diagonal walks $S = (x_1 + \overline{x_1})(x_2 + \overline{x_2})$) we have a linear relationship between these quantities. In other cases (like cartesian walks $S = x_1 + \overline{x}_1 + x_2 + \overline{x}_2$) we don't have such a linear relationship but we have a rational one, that is G is a rational function of G_1, G_2 and $G_{1,2}$. The aim of the followin sections is to investigate when such relationships exist, when se have a relationship of that type and whether we have other cases.

The work can be split into several cases - walks that never leave the origin and ones that leave but never return to the origin. In the first case the relationship is trivially $G - G_1 - G_2 + G_{1,2} = 0$ because $Q = 1$ itself and so $G = 1$. In the second case Q may be more complicated but again $G = 1$ trivially since no walks end at the origin. This case satisfies the same relationship. However, once we move beyond this stage we need to consider the problem more carefully.

12 Simple cases: Walks never leaving the origin and walks leaving the origin but never returning

Here are all the easy cases of step sets that lead to walks that either don't leave the origin or leave it and never come back. There are 80 such sets among the 256 possible ones. Of course we could use symmetries to eliminate cases but for now this was an easier thing to do.



There are $2^5 = 32$ types of walks that don't leave the origin. Those are walks with steps among the set {NW,W,SW,S,SE}. Next we look at the walks that leave the origin but never come back. There is the case of $S = \{NE\}$ that never comes back to any axis. Let's look at those walks that never come back to the x_1 axis. We have some steps that go in the positive x_2 direction, that is one of {N, NE, E} together with some steps from {NW, W, SW}. If we were to include any other steps we would come back to the x_1 axis. If we only have a N step, we have 8 new cases, if we have only a NE step we have 3 new cases and if we have both, we have 4 new cases. This gives a total of 15 new cases. There's an equivalent number of walks that leave the x_2 axis and never come back to it. The running total is $32 + 1 + 15 + 15 = 63$.

To finish off we have 17 cases of walks that leave the origin, eventually interact with both axes but never come back to the origin. For this we must only include steps in such a way that we cannot pick a selection of steps that will decrease both the x_1 and x_2 coordinates.

We have to pick a subset of steps from {N, NE, E} so the only other steps allowed are NW and SE. The new cases coming from {N, NE, E}, {N,E}, {N, NE }, {NE, E}, {N}, {NE}, {E} are 4, 4, 2, 2, 1, 2, 2 respectively.

Before we consider walks that come back to the origin we'll look at the behavior Q depending on whether S factors in some nice way or not.

13 Simple case: Splitting S into a summation

First we consider a simple linear factoring of $S = S_1(x_1) + S_2(x_2)$. First we have that $D_1 S_2 = S_2$ since D_1 doesn't interact with the x_2 variable. Similarly $D_2 S_1 = S_1$. Then

$$DS(1) = D_1 D_2 S = D_1 D_2 (S_1 + S_2) = D_1 S_1 + D_2 S_2$$

Furthermore

$$D_1 S_1 D_2 S_2 = D_2 S_2 D_1 S_1$$

so they commute. By induction we can write

$$(DS)^n = (D_1 S_1 + D_2 S_2)^n = \sum_{i=0}^{n} \binom{n}{i} (D_1 S_1)^i (D_2 S_2)^{n-i}$$

This means that we have

$$\begin{aligned} Q &= \sum_{k=0}^{\infty} t^k (DS)^n(1) \\ &= \sum_{k=0}^{\infty} t^k \sum_{i=0}^{n} \binom{n}{i} (D_1 S_1)^i (1)(D_2 S_2)^{n-i}(1) \\ &= \left(\sum_{k=0}^{\infty} t^k (D_1 S_1)^k (1) \right) \left(\sum_{k=0}^{\infty} t^k (D_2 S_2)^k (1) \right) \\ &= Q_1(x_1, a_1) Q_2(x_2, a_2) \end{aligned}$$

That is Q factors into two functions! The splitting of the a_i comes from the fact that the a_1 label comes from D_1 and a_2 from D_2. This reduces the problem to two one-dimensional problems which is always welcome. Here's something neat

$$\begin{aligned} Q = Q_1 Q_2, \quad \widehat{Q} = \widehat{Q}_1 Q_2 \\ \breve{Q} = Q_1 \breve{Q}_2, \quad \breve{\widehat{Q}} = \widehat{Q}_1 \breve{Q}_2 \end{aligned}$$

Therefore

$$Q - \widehat{Q} - \breve{Q} + \breve{\widehat{Q}} = (Q_1 - \widehat{Q}_1)(Q_2 - \breve{Q}_2)$$

The concept generalizes to higher dimensions. In d dimensions suppose that S splits as a sum into k parts $S = S_1 + \ldots S_k$ no two of which share variables. Then the problem boils down

to solving k lower dimensional problems, renaming variables appropriately and multiplying the result

$$Q = Q_1 \dots Q_k$$

Solving lower-dimensional functional equations for Q leads to solutions to infinite families of higher-dimensional problems with almost no effort. This works with or without labelling the interactions with the hyperplanes $x_i = 0$. This provides a way of finding higher dimensional numbers of restricted walks.

Unfortunately there aren't that many interesting examples of this splitting in two dimensions. Since we've already solved the $d = 1$ problem completely we only need to delineate all the cases. We have $S_1(x), S_2(x) \in \{0, x, \overline{x}+x, \overline{x}\}$. Out of 16 possible models like this, 4 never leave the origin, 5 leave the origin but never come back. These have already been considered. Out of the 7 remaining ones that return to the origin, 4 are really one-dimensional, 2 are interesting two-dimensional ones that are equivalent to Dyck paths, with one truly new example - walks with steps $\{N, S, E, W\}$. Solving these will bring the tally of solved models to 87.

14 Simple case: Factoring S into a product

The next case is of factoring S as $S = S_1(x_1)S_2(x_2)$. We have

$$(DS)^k = (D_1 D_2 S_1 S_2)^k = (D_1 S_1)^k (D_2 S_2)^k$$

so we can compute Q directly as

$$Q = \sum_{k=0}^{\infty} t^k (DS)^k = \sum_{k=0}^{\infty} t^k (D_1 S_1)^k (D_2 S_2)^k = Q_1(x_1, a_1) \odot Q_2(x_2, a_2)$$

where $\odot$ denotes the Hadamard product of two functions. In other words, if we want to look at the generating function for quarter plane walks of length k it is enough to look at the product of two generating functions for one-dimensional walks of length k. Similarly to the previous section we can write

$$Q = Q_1 \odot Q_2, \quad \widehat{Q} = \widehat{Q}_1 \odot Q_2$$
$$\breve{Q} = Q_1 \odot \breve{Q}_2, \quad \breve{\widehat{Q}} = \widehat{Q}_1 \odot \breve{Q}_2$$

Therefore

$$Q - \widehat{Q} - \breve{Q} + \breve{\widehat{Q}} = (Q_1 - \widehat{Q}_1) \odot (Q_2 - \breve{Q}_2)$$

Factoring also generalizes to d dimensions. If S factors into a product of k factors $S_1, \dots, S_k$ where no two of which share any variables then the problem of finding Q boils down to solving k lower-dimensional problems for functions $Q_1, \dots, Q_k$, naming variables appropriately and taking the Hadamard product of the functions. That is

$$Q = Q_1 \odot \dots \odot Q_k$$

Here's one quick application. If there are $q_{k,n}$ walks of length n counted by the function Q_k, then there are $q_{1,n} \dots q_{k,n}$ walks of length k counted by Q. This should make giving asymptotics for higher dimensional problems easier.

15 Interesting 2D cases: Full walks

16 Conjectured linear and rational relationships depending on $S_{1,2}^{-1,-1}$

17 Further analysis of the 2D equation

We can perform coefficient extractions of the master equation to get the following working forms

$$(1-tS)Q-(c_1-t\overline{x_1}S_1^{-1})Q_1^0-(c_2-t\overline{x_2}S_2^{-1})Q_2^0+(c_1c_2-t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0}=\overline{a_1a_2} \quad (11)$$

$$(1-ta_1S_1^0)Q_1^0-(c_2-ta_1\overline{x_2}S_{1,2}^{0,-1})Q_{1,2}^{0,0}-ta_1(S_1^{-1}Q_1^1-\overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0})=\overline{a_2} \quad (12)$$

$$(1-ta_2S_2^0)Q_2^0-(c_1-ta_2\overline{x_1}S_{1,2}^{-1,0})Q_{1,2}^{0,0}-ta_2(S_2^{-1}Q_2^1-\overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1})=\overline{a_1} \quad (13)$$

$$Q_{1,2}^{0,0}-ta_1a_2(S_{1,2}^{-1,-1}Q_{1,2}^{1,1}+S_{1,2}^{-1,0}Q_{1,2}^{1,0}+S_{1,2}^{0,-1}Q_{1,2}^{0,-1})=1 \quad (14)$$

There might be some operator form that will allow us to write these out exactly.

18 Diagonal steps: Deriving the Extraction-Ready Functional Equations

Let's take the case of the diagonal steps with a finite group G of order 4. Let $\bullet : x_1 \mapsto \overline{x_1}$ and $* : x_2 \mapsto \overline{x_2}$. We have

$$K^\bullet = K^* = K, \quad K_2^\bullet = K_2, \quad K_1^* = K_1$$

while the others change. We can also solve when $K = 0$ for x_1. There are two roots

$$\sigma_1 = \frac{1-\sqrt{1-4t^2(S_1^{-1})^2}}{2tS_1^{-1}}, \quad \overline{\sigma_1} = \frac{1+\sqrt{1-4t^2(S_1^{-1})^2}}{2tS_1^{-1}}$$

with the extra property that

$$\sigma_1 + \overline{\sigma_1} = \frac{1}{tS_1^{-1}}.$$

We can take the master equation and apply $*$ to it

$$-KQ + K_1Q_1^0 + K_2Q_2^0 - K_{1,2}Q_{1,2}^{0,0} + \alpha = 0$$
$$-KQ^* + K_1Q_1^{0,*} + K_2^*Q_2^0 - K_{1,2}^*Q_{1,2}^{0,0} + \alpha = 0$$

From here on we will input $x_1 = \sigma_1$ to rid ourselves of the kernel K. There will be an overloading of notation

$$K_1Q_1^0 + K_2Q_2^0 - K_{1,2}Q_{1,2}^{0,0} + \alpha = 0$$
$$K_1Q_1^{0,*} + K_2^*Q_2^0 - K_{1,2}^*Q_{1,2}^{0,0} + \alpha = 0$$

We can multiply both equation by something to clear Q_2^0 because it involves σ_1 while the other term involves only x_2. We have

$$K_1K_2^*Q_1^0 + K_2^*K_2Q_2^0 - K_2^*K_{1,2}Q_{1,2}^{0,0} + \alpha K_2^* = 0$$
$$K_1K_2Q_1^{0,*} + K_2K_2^*Q_2^0 - K_2K_{1,2}^*Q_{1,2}^{0,0} + \alpha K_2 = 0$$

We subtract these two equations to obtain

$$K_1(K_2^*Q_1 - K_2Q_1^*) - (K_2^*K_{1,2} - K_2K_{1,2}^*)Q_{1,2} + \alpha(K_2^* - K_2) = 0$$
$$K_2^*Q_1 - K_2Q_1^* - \frac{K_2^*K_{1,2} - K_2K_{1,2}^*}{K_1}Q_{1,2} + \alpha\frac{K_2^* - K_2}{K_1} = 0$$

First we have

$$\begin{aligned} K_2^* - K_2 &= t(\overline{x_2} - x_2)S_2^{-1}(\sigma_1) \\ &= t(\overline{x_2} - x_2)(\overline{\sigma_1} + \sigma_1) \\ &= \frac{\overline{x_2} - x_2}{\overline{x_2} + x_2} \end{aligned}$$

We have

$$\begin{aligned} K_2^*K_{1,2} - K_2K_{1,2}^* &= (c_2 - tx_2S_2^{-1})(c_1c_2 - t\overline{\sigma_1 x_2}) - (c_2 - t\overline{x_2}S_2^{-1})(c_1c_2 - t\overline{\sigma_1}x_2) \\ &= c_1c_2t(\overline{x_2} - x_2)S_2^{-1} - c_2t\overline{\sigma_1}(\overline{x_2} - x_2) \\ &= c_2t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) \end{aligned}$$

If we let $v = \overline{x_2} + x_2$ we find

$$\sigma_1 = \frac{1 - \sqrt{1 - 4t^2v^2}}{2tv}, \quad \overline{\sigma_1} = \frac{1 + \sqrt{1 - 4t^2v^2}}{2tv}$$

We have

$$\begin{aligned} (c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) &= \frac{c1}{tv} - \frac{1 + \sqrt{1 - 4t^2v^2}}{2tv} \\ &= \frac{1}{\overline{x_2} + x_2}\left(-\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2v^2}}{2}\right) \end{aligned}$$

Also

$$\begin{aligned} K_1 &= c_1 - tv\overline{\sigma_1} \\ &= c_1 - tv\frac{1 + \sqrt{1 - 4t^2v^2}}{2tv} \\ &= 1 - \overline{a_1} - \frac{1 + \sqrt{1 - 4t^2v^2}}{2} \\ &= -\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2v^2}}{2} \end{aligned}$$

Thus we finally have

$$\begin{aligned}\frac{1}{K_1}(K_2^* K_{1,2} - K_2 K_{1,2}^*) &= -\frac{c_2 t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1})}{c_1 - t\overline{\sigma_1}v} \\ &= -\frac{c_2 t(\overline{x_2} - x_2)(-1)}{tv} \\ &= \frac{c_2(\overline{x_2} - x_2)}{\overline{x_2} + x_2}\end{aligned}$$

Thus the equation will now read as

$$K_2^* Q_1 - K_2 Q_1^* - \frac{c_2(\overline{x_2} - x_2)}{\overline{x_2} + x_2} Q_{1,2} + \alpha \frac{\overline{x_2} - x_2}{(\overline{x_2} + x_2)K_1} = 0$$

We just need to take care of the remaining terms. We'll do this by multiplying by $\overline{x_2} + x_2$. We have

$$\begin{aligned}R^* &= (\overline{x_2} + x_2)K_2^* = c_2(\overline{x_2} + x_2) - x_2 = c_2\overline{x_2} - \overline{a_2}x_2 \\ R &= (\overline{x_2} + x_2)K_2 = c_2(\overline{x_2} + x_2) - \overline{x_2} = c_2 x_2 - \overline{a_2 x_2}\end{aligned}$$

We have a couple of relations here

$$\begin{aligned}R^* - R &= v(K_2^* - K_2) = \overline{x_2} - x_2 \\ \overline{x_2}R^* - x_2 R &= \overline{x_2}c_2 v - 1 - x_2 c_2 v + 1 \\ &= c_2 v(\overline{x_2} - x_2)\end{aligned}$$

The equation becomes

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^0(x_2) + (\overline{a_2 x_2} - c_2 x_2)Q_1^0(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + (\overline{x_2} - x_2)\overline{K}_1\alpha = 0$$

and it's ready for extraction. Let $\delta_0 = \overline{a_1}(\overline{x_2} - x_2)\overline{K}_1$. We re-write the equation as

$$R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 = 0$$

We'll use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x_1 x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1 a_2}$$

to further our work here. For diagonal steps it is

$$(1 - tS)Q - (c_1 - tv\overline{x_1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x_1 x_2})Q_{1,2}^{0,0} = \overline{a_1 a_2}$$

extracting the $[x_1^0]$ coefficient yields

$$Q_1^0 = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1$$

We can extract higher coefficients too. Let $k \geqslant 1$. Then the $[x_2^k]$ coefficient is

$$\begin{aligned}0 &= Q_1^k - tS_1^{-1}Q_1^{k+1} - tS_1^0Q_1^k - tS_1^1Q_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1}Q_{1,2}^{k,0} + S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) \\ &= Q_1^k - tvQ_1^{k+1} - tvQ_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})\end{aligned}$$

We can re-write the last one in a form more convenient for the next few steps

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2Q_{1,2}^{k,0} - t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})$$

We'll use induction to prove a nice formula but for that we'll first need to establish another base case because of this splitting of cases for the coefficient extraction. We have

$$\begin{aligned}Q_1^0 &= \overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1 \\ Q_1^{0,*} &= \overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^{1,*}\end{aligned}$$

Combining them with equation (18) gives

$$\begin{aligned}0 &= R^*Q_1^0 - RQ_1^{0,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= R^*(\overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1) - R(\overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^{1,*}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + (R^* - R)(\overline{a_2} + c_2Q_{1,2}^{0,0}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + \overline{a_2}(\overline{x_2} - x_2) + \overline{a_2}\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + \overline{a_2}((\overline{x_2} - x_2) + \delta_0)\end{aligned}$$

So we can write the equation

$$R^*Q_1^1 - RQ_1^{1,*} + \overline{a_2}\delta_1 = 0$$

where

$$\delta_1 = \frac{(\overline{x_2} - x_2) + \delta_0}{a_1tv}$$

This equation together with equation (18) form the base cases for the following lemma

Lemma 5. Let $m \geqslant 0$ be an integer. Then

$$R^*Q_1^m - RQ_1^{m,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0 \tag{15}$$

where δ_m is not a function of a_2.

Proof. We'll show this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$ since we cannot have any walks end on coordinates $(1, 0)$. Now we assume that the formula is true for all cases up to and including m. Then

we use formula (18)

$$\begin{aligned}
&R^*Q_1^m - RQ_1^{m,*}\\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(R^*Q_1^{m-1} - RQ_1^{m-1,*})\\
&+ c_2(R^* - R)Q_{1,2}^{m,0} - t(\overline{x_2}R^* - x_2R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0})\\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(c_2(\overline{x_2} - x_2)Q_{1,2}^{m-1,0} - \overline{a_2}\delta_{m-1})\\
&+ c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x_2} - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0})\\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x_2} - x_2)Q_{1,2}^{m+1,0}) - \overline{a_2}tv\delta_{m-1})
\end{aligned}$$

Combining this with equation (15) leads to

$$\begin{aligned}
0 &= tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) - c_2tv(\overline{x_2} - x_2)Q_{1,2}^{m+1,0} - \overline{a_2}tv\delta_{m-1} + \overline{a_2}\delta_m\\
0 &= R^*Q_1^{m+1} - RQ_1^{m+1,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{m+1,0} + \overline{a_2}\delta_{m+1}
\end{aligned}$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1}$$

This function δ_{m+1} is also not dependent on a_2 and this completes the proof! □

There is an equivalent statement for Q_2^n. We can expand R^* and R to a form we can use

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^m(x_2) - (c_2x_2 - \overline{a_2x_2})Q_1^m(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0$$

We start by extracting the $[x_2^1]$ coefficient

$$\begin{aligned}
&c_2Q_{1,2}^{m,2} - \overline{a_2}Q_{1,2}^{m,0} - c_2Q_{1,2}^{m,0} + c_2Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0\\
&c_2Q_{1,2}^{m,2} - \overline{a_2}Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0
\end{aligned}$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geqslant 2$

$$c_2Q_{1,2}^{m,n+2} - \overline{a_2}Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a_2}\delta_m = 0$$

This formula also contains the $n = 0$ case so we treat them together. We can set $a_2 = 1$ to get

$$-\breve{Q}_{1,2}^{m,n} + [x_2^{n+1}]\delta_m = 0$$

Substituting this back into the above equation gives

$$b_2Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n}$$

An equivalent formula can be derived if the process is followed through the use of x_1 instead of x_2 in the analysis.

Corollary 6. For integers $m, n \geqslant 0$ we have the following relations

$$\begin{aligned}
b_1Q_{1,2}^{m+2,n} &= Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n}\\
b_2Q_{1,2}^{m,n+2} &= Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n}
\end{aligned}$$

We can sum up those equations to obtain functional equations

Corollary 7. For integers $m \geqslant 0$ we have the following relations

$$\begin{aligned}
Q_1^m - \widehat{Q}_1^m &= b_1 Q_1^{m+2} \\
\breve{Q}_1^m - \breve{\widehat{Q}}_1^m &= b_1 \breve{Q}_1^{m+2} \\
x_1^2(Q_2^n - \widehat{Q}_2^n) &= b_1(Q_2^n - Q_{1,2}^{1,n} x_1 - Q_{1,2}^{0,n}) \\
x_1^2(\breve{Q}_2^n - \breve{\widehat{Q}}_2^n) &= b_1(\breve{Q}_2^n - \breve{Q}_{1,2}^{1,n} x_1 - \breve{Q}_{1,2}^{0,n}) \\
x_1^2(Q - \widehat{Q}) &= b_1(Q - x_1 Q_1^1 - Q_1^0) \\
x_1^2(\breve{Q} - \breve{\widehat{Q}}) &= b_1(\breve{Q} - x_1 \breve{Q}_1^1 - \breve{Q}_1^0)
\end{aligned}$$

Corollary 8. For integers $n \geqslant 0$ we have the following relations

$$\begin{aligned}
Q_2^n - \breve{Q}_2^n &= b_2 Q_2^{n+2} \\
\widehat{Q}_2^n - \breve{\widehat{Q}}_2^n &= b_2 \widehat{Q}_2^{n+2} \\
x_2^2(Q_1^m - \breve{Q}_1^m) &= b_2(Q_1^m - Q_{1,2}^{m,1} x_1 - Q_{1,2}^{m,0}) \\
x_2^2(\widehat{Q}_1^m - \breve{\widehat{Q}}_1^m) &= b_2(\widehat{Q}_1^m - \widehat{Q}_{1,2}^{m,1} x_1 - \widehat{Q}_{1,2}^{m,0}) \\
x_2^2(Q - \breve{Q}) &= b_2(Q - x_2 Q_2^1 - Q_2^0) \\
x_2^2(\widehat{Q} - \breve{\widehat{Q}}) &= b_2(\widehat{Q} - x_2 \widehat{Q}_2^1 - \widehat{Q}_2^0)
\end{aligned}$$

Every other equation is obtained from the previous one by either setting $a_1 = 1$ or $a_2 = 1$. Here we can hope to set some telescoping sum type of thing in order to say more about $Q_{1,2}^{0,0}$

Now we can move onto the main result about the diagonal walks and it is that the function for labelled walks ending at $(0, 0)$ is connected to the functions for semi-labelled and unlabelled walks endind at $(0, 0)$. We can use the master equation to extract the $[x_1^0 x_2^0]$ and $[x_1^1 x_2^1]$ coefficients yielding the following two relations

$$\begin{aligned}
Q_{1,2}^{0,0} &= 1 + t a_1 a_2 Q_{1,2}^{1,1} \\
Q_{1,2}^{1,1} &= t(Q_{1,2}^{0,0} + Q_{1,2}^{2,0} + Q_{1,2}^{0,2} + Q_{1,2}^{2,2})
\end{aligned}$$

We also have

$$\begin{aligned}
b_1 b_2 Q_{1,2}^{2,2} &= b_1(Q_{1,2}^{2,0} - \breve{Q}_{1,2}^{2,0}) \\
&= b_1 Q_{1,2}^{2,0} - b_1 \breve{Q}_{1,2}^{2,0} \\
&= Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} - \breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0}
\end{aligned}$$

together with

$$\begin{aligned}
b_1 b_2 Q_{1,2}^{2,0} &= b_2 Q_{1,2}^{0,0} - b_2 \widehat{Q}_{1,2}^{0,0} \\
b_1 b_2 Q_{1,2}^{0,2} &= b_1 Q_{1,2}^{0,0} - b_1 \breve{Q}_{1,2}^{0,0}
\end{aligned}$$

We we have

$$\begin{aligned} b_1b_2Q^{1,1}_{1,2} &= t(b_1b_2Q^{0,0}_{1,2} + b_1b_2Q^{2,0}_{1,2} + b_1b_2Q^{0,2}_{1,2} + b_1b_2Q^{2,2}_{1,2}) \\ &= t(b_1b_2Q^{0,0}_{1,2} + b_2Q^{0,0}_{1,2} - b_2\widehat{Q}^{0,0}_{1,2} + b_1Q^{0,0}_{1,2} - b_1\breve{Q}^{0,0}_{1,2} + Q^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} - \breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \\ &= t((b_1b_2 + b_1 + b_2 + 1)Q^{0,0}_{1,2} - (b_2+1)\widehat{Q}^{0,0}_{1,2} - (b_1+1)\breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \\ &= t(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \end{aligned}$$

We can finally write

$$\begin{aligned} b_1b_2Q^{0,0}_{1,2} &= b_1b_2 + ta_1a_2b_1b_2Q^{1,1}_{1,2} \\ &= b_1b_2 + t^2a_1a_2(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \end{aligned}$$

Let $G(b_1, b_2)$ stand for $Q(0, 0, b_1, b_2)$ and G_i has $b_i = 0$. Then we can write it all as

$$b_1b_2G = b_1b_2 + t^2a_1a_2(a_1a_2G - a_2G_1 - a_1G_2 + G_{1,2})$$

or written as a linear combination

$$(b_1b_2 - t^2a_1^2a_2^2)G + t^2a_1a_2^2G_1 + t^2a_1^2a_2G_2 - t^2a_1a_2G_{1,2} = b_1b_2$$

We can divide out by a_1a_2 to have the nicer expression

$$(c_1c_2 - a_1a_2t^2)G + a_2t^2G_1 + a_1t^2G_2 - t^2G_{1,2} = c_1c_2$$

We can even write

$$G = \frac{(c_1c_2 - t^2(a_2G_1 + a_1G_2 - G_{1,2})}{c_1c_2 - a_1a_2t^2}$$